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Strain-tunable ferromagnetic to antiferromagnetic and half-metallic to ferroelectric phase transitions in a 2D bimetallic oxyhalide **FREE**

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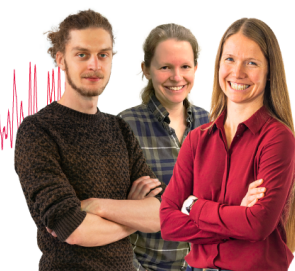
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ABSTRACT

The coexistence and tunability of ferroic orders in two-dimensional materials hold great promise for next-generation spintronic and memory devices. Here, based on first-principles calculations, we demonstrate that the interplay of chemical substitution and strain engineering enables controlled phase transitions among ferromagnetic, antiferromagnetic, half-metallic, and ferroelectric states in a two-dimensional bimetallic oxyhalide monolayer NbMnO₂Cl₄. The pristine NbMnO₂Cl₄ monolayer exhibits intrinsic half-metallic ferromagnetism, with a Curie temperature of 285 K and magnetic anisotropy energy of 225 μ eV per Mn atom. The compressive biaxial strain triggers a sequential transition from half-metallic ferromagnetism to ferroelectric–ferromagnetic and subsequently to ferroelectric–antiferromagnetic states. Conversely, tensile biaxial strain drives a direct transition from half-metallic ferromagnetism to a ferroelectric–antiferromagnetic phase. Spin–lattice coupling plays a crucial role in these phase transitions. Moreover, the Curie temperature increases to 540 K at a tensile strain of 6%, and the magnetic anisotropy energy reaches 670 μ eV per Mn atom at a compressive strain of –8%. These findings offer an approach for tailoring multiple ferroic orders in two-dimensional materials through the synergistic effects of chemical doping and mechanical strain.

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The van der Waals niobium metal oxide dihalides NbOX₂ (X = Cl, Br, and I) have recently attracted significant attention in condensed matter physics due to their remarkable in-plane anisotropy and giant second-order nonlinear optical response.^{1–3} Both theoretical and experimental studies have demonstrated that two-dimensional (2D) NbOX₂ compounds are intrinsic ferroelectrics (FEs) exhibiting a strong second-harmonic generation response, primarily attributed to a Peierls distortion driven by in-plane anisotropic bonding.^{4–7} Notably, the spontaneous polarization in monolayer NbOCl₂ is stable above room temperature,⁶ highlighting its potential for applications in nonvolatile FE nanomemory devices.

In addition to the FE behavior, NbOX₂ monolayers also exhibit unconventional electronic characteristics associated with the partially filled Nb 4d orbitals.^{7,8} Each Nb⁴⁺ cation possesses one unpaired d electron, half filling the d_{xy} orbital, which would typically give rise to a local magnetic moment. However, due to the dimerization between neighboring Nb atoms, orbital overlap occurs between adjacent d_{xy} orbitals, leading to the quenching of the local magnetic moments,

limiting the potential of NbOX₂ for spin-based applications. Therefore, it is of great interest to explore possible strategies to break the Nb–Nb dimerization and stabilize a magnetic state.

Strain engineering offers a direct and effective route to destabilize the Nb–Nb dimerization, thereby suppressing orbital overlap and restoring local spin moment. Theoretical calculations have demonstrated that applying uniaxial strain can induce a phase transition from a nonmagnetic (NM) to an antiferromagnetic (AFM) state in monolayer NbOCl₂.⁸ In addition, external strain can also give rise to anisotropic and enhanced phonon responses,⁹ FE phase transitions,^{10,11} and significantly enhanced linear electro-optic responses¹¹ and second-harmonic generation¹⁰ in this system.

Beyond strain engineering, chemical substitution with magnetic transition-metal elements could provide an alternative strategy to activate magnetism in the NbOX₂ monolayer. As a prototypical magnetic transition metal with multiple oxidation states and high-spin electronic configurations, Mn has been widely employed to induce robust ferromagnetic (FM) ordering in otherwise NM hosts.^{12–16} Examples include

Mn doping induces room temperature, robust FM ordering in transition-metal dichalcogenides,¹⁴ and half-metallic ferromagnetic (HM-FM) behavior in FE In_2Se_3 monolayer.¹³ Motivated by these insights, we constructed NbMnO_2X_4 ($\text{X}=\text{F}$, Cl , and Br) monolayers by substituting Mn for Nb to investigate how chemical doping influences the structural, ferroelectric, and magnetic properties of the system as well as their evolution under external strain.

Our first-principles calculations reveal that Mn substitution stabilizes a HM-FM ground state in monolayer NbOCl_2 , with a magnetic anisotropy energy (MAE) of $225 \mu\text{eV}$ per Mn atom and a high Curie temperature of 285 K. Under compressive biaxial strain, this system undergoes two successive transitions: first to a FE-FM state and then to a FE-AFM state. Under tensile strain, it directly changes to a FE-AFM phase. The Curie temperature increases to 540 K at 6% tensile strain, and the MAE reaches $670 \mu\text{eV}$ per Mn atom at -8% compressive strain.

We first constructed NbMnO_2X_4 monolayers by substituting Mn for Nb. Phonon calculations revealed that only monolayer $\text{NbMnO}_2\text{Cl}_4$ is dynamically stable [Fig. S1(a)]. AIMD simulations further confirmed its thermal stability [Fig. S1(b)]. As illustrated in Fig. 1(a), Mn and Nb atoms reside in edge-sharing octahedral chains, which are interconnected via O atoms along the a direction. Nb atoms exhibit off-center displacements along the b direction, shifting toward the Cl vertex within the $[\text{NbO}_2\text{Cl}_4]$ octahedron, leading to the formation of Mn-Nb dimers (i.e., $d_1 < d_2$), as depicted in Fig. 1(b). The Nb displacement along the b direction is $0.24 \text{ \AA}$, which results in the separation of positive and negative charge centers, thereby inducing spontaneous polarization, as shown in Fig. 1(c). The optimized lattice constants are $a = 3.7 \text{ \AA}$ and $b = 6.6 \text{ \AA}$, and the monolayer belongs to the $Pmm2$ space group.

The calculated cohesive energy is -2.93 eV/atom , comparable to those of NbOX_2 monolayers (Table S1), implying feasibility for synthesis. The calculated elastic constants— $C_{11} = 37.13 \text{ N/m}$, $C_{22} = 16.64 \text{ N/m}$,

$C_{12} = -0.70 \text{ N/m}$, and $C_{66} = 4.27 \text{ N/m}$ —satisfy the Born criteria for 2D materials,¹⁷ confirming mechanical stability of monolayer $\text{NbMnO}_2\text{Cl}_4$. Figure S2(a) shows the corresponding 2D Young's moduli of monolayer $\text{NbMnO}_2\text{Cl}_4$: $E_x = 37.10 \text{ N/m}$ and $E_y = 16.63 \text{ N/m}$ along the x and y directions, respectively. Although these values are smaller than those of NbOX_2 monolayers, e.g., $E_x = 80.3 \text{ N/m}$ and $E_y = 65.9 \text{ N/m}$ in NbOI_2 ,⁸ monolayer $\text{NbMnO}_2\text{Cl}_4$ uniquely exhibits negative Poisson's ratios, a mechanical behavior not found in its NbOX_2 counterparts. Specifically, the Poisson's ratios are -0.042 and -0.019 along the x and y directions, respectively [Fig. S2(b)], comparable to other 2D materials.^{18–20} Evidence for the negative Poisson's ratio of monolayer $\text{NbMnO}_2\text{Cl}_4$ is further provided by strain–stress curves (Fig. S3), together with a detailed analysis of its microscopic origin.

Considering the Hubbard U correction significantly influences crystal structures with partially occupied d orbitals,²¹ we analyzed its effects systematically. As shown in Fig. 1(b), increasing the effective Hubbard parameter ($U_{\text{eff}} = U - J$) for Mn atoms results in lattice expansions along both the a and b axes, with the lattice constant along the a axis increasing faster, particularly for U_{eff} values between 1 and 2 eV. This trend weakens Mn–Nb dimerization and reduces polar distortion, evident from increased Mn–Nb bond lengths (d_1 , d_2) and Mn–Cl–Nb bond angle [α , Fig. 1(b)]. At $U_{\text{eff}} \geq 4 \text{ eV}$, the polar distortion completely disappears ($\text{PD} = 0$), transitioning the structure to a centrosymmetric, paraelectric phase with $Pmmm$ symmetry [Fig. 1(d)]. In contrast, variations of the Hubbard U applied to Nb atoms show minimal impact on the structure (Fig. S4).

Given that a Hubbard U of 5 eV is commonly employed for Mn atoms,^{22–26} we adopt $U_{\text{eff}} = 5 \text{ eV}$ for Mn $3d$ orbitals and $U_{\text{eff}} = 3 \text{ eV}$ for Nb $4d$ orbitals in subsequent calculations. The spin-polarized band structures presented in Figs. 2(a)–2(c) and the corresponding spin-resolved projected density of states (PDOS, Fig. S5) confirm half-metallicity in monolayer $\text{NbMnO}_2\text{Cl}_4$, characterized by a metallic spin-up channel and a semiconducting spin-down channel with a spin

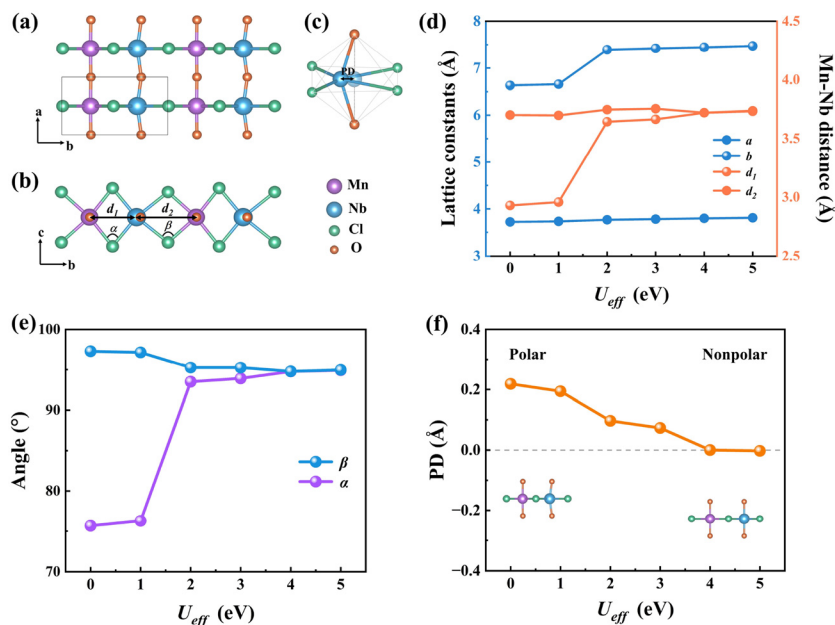


FIG. 1. (a) Top and (b) side views of the crystal structure of monolayer $\text{NbMnO}_2\text{Cl}_4$, optimized using the standard PBE method. This structure belongs to space group $Pmm2$ and point group C_{2v} . Purple, blue, green, and orange spheres denote Mn, Nb, Cl, and O atoms, respectively. (c) The $[\text{NbO}_2\text{Cl}_4]$ octahedra unit, where Nb atom exhibits non-zero off-center polar displacement (PD). Dependence of structural parameters on the effective Hubbard parameter U_{eff} for Mn: (d) lattice constants a and b , and Mn–Nb bond lengths d_1 and d_2 ; (e) $\angle \text{Mn–Cl–Nb}$ and $\angle \text{Nb–Cl–Mn}$ bond angles α and β [as defined in (b)]; and (f) polar displacement (PD) [as defined in (c)].

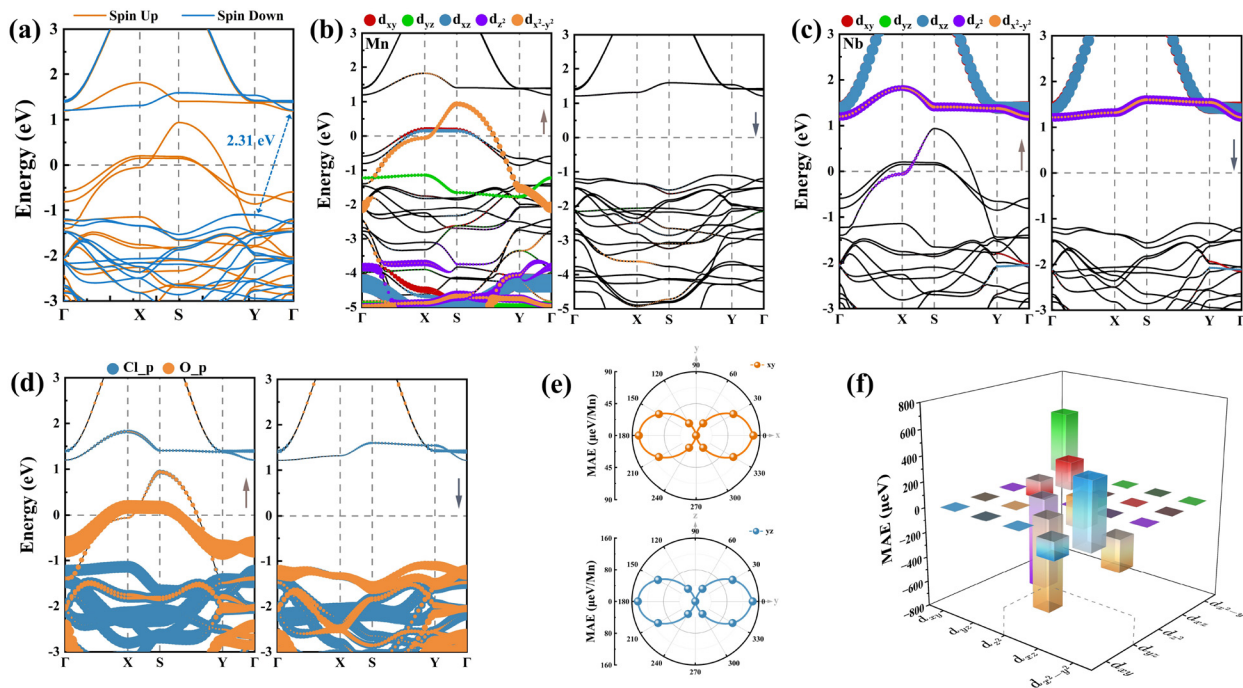


FIG. 2. (a) Spin-polarized electronic band structure of monolayer NbMnO₂Cl₄ with $U = 5$ and 3 eV for Mn and Nb atoms, respectively. Orange and blue lines indicate spin-up and spin-down states, respectively. (b)–(d) Corresponding band structures projected onto Mn-3d, Nb-4d, and Cl/O-p orbitals. Horizontal dashed lines represent the Fermi level. (e) Angular dependence of the MAE in units μ eV per Mn atom in the xy (upper panel) and yz (lower panel) planes. (f) Orbital-resolved MAE, highlighting contributions from Mn-3d orbitals.

gap of 2.31 eV, as clearly shown in Fig. 2(a). Three bands crossing the Fermi level primarily originate from Mn $d_{x^2-y^2}$, d_{xy} , and d_{xz} orbitals as well as O p orbitals [Figs. 2(a) and 2(d)], contributing to the metallic behavior in the spin-up channel. In the semiconducting spin-down channel, the highest occupied and lowest unoccupied states mainly derive from O p and Nb d_{z^2} orbitals, respectively [Figs. 2(c) and 2(d)]. Spin-orbit coupling shows negligible impact on the electronic structure and is, thus, omitted in subsequent calculations (Fig. S6).

To determine the magnetic ground state of monolayer NbMnO₂Cl₄, four different magnetic configurations were examined in a $2 \times 2 \times 1$ supercell: one FM and three AFM (AFMa, AFMb, and AFMc) states (Fig. S7). Among them, the FM configuration exhibits the lowest total energy, indicating ferromagnetic ground state. The calculated magnetic moments are $4.3 \mu_B$ for Mn and $0.05 \mu_B$ for Nb, which are consistent with the high-spin Mn³⁺ ($3d^4$) and nonmagnetic Nb⁵⁺ ($4d^0$) valence states.

The MAE was calculated to determine the easy axis of magnetization [Fig. 2(e)]. The results reveal that monolayer NbMnO₂Cl₄ favors an out-of-plane easy axis of magnetization. The angular dependence of MAE in the xy and yz planes reveals strong magnetic anisotropy, with the maximum MAE reaching $225 \mu\text{eV}$ per Mn atom, which is significantly larger than that of bulk Fe ($1.4 \mu\text{eV}/\text{atom}$) and Ni ($3 \mu\text{eV}/\text{atom}$) as well as those of 2D FeP₄ ($72.55 \mu\text{eV}/\text{Fe}$),²⁷ 2D h -MnN ($43 \mu\text{eV}/\text{Mn}$),²⁸ 2D Cr₂BN ($103 \mu\text{eV}/\text{Cr}$),²⁹ CrSnTe₃ ($69 \mu\text{eV}/\text{Cr}$),³⁰ and CrCl₃ ($69 \mu\text{eV}/\text{Cr}$).³¹ Orbital-resolved MAE analysis [Fig. 2(f)] further indicates that the dominant contributions originate from spin-orbit

coupling between the d_{z^2} and d_{yz} orbitals as well as between the d_{xy} and $d_{x^2-y^2}$ orbitals.

The T_c of monolayer NbMnO₂Cl₄ was estimated by performing MC simulations based on the classical Heisenberg model. The spin Hamiltonian is defined as

$$H = -J_1 \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j - J_2 \sum_{\langle\langle i,k \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_k,$$

where J_1 and J_2 represent the nearest-neighbor and next-nearest-neighbor magnetic exchange parameters, respectively. The summations i, j , and k run over all the Mn sites. $\mathbf{S}$ denotes the formal magnetic moment of Mn ions. Positive and negative J values correspond to FM and AFM couplings, respectively. From energy-mapping analyses based on the considered magnetic configurations, we obtained $J_1 = 91$ meV and $J_2 = 2$ meV. The estimated T_c is 285 K (Fig. S8), which is close to room temperature and significantly higher than those of other typical 2D FM materials, such as CrI₃ (45 K)³² and Cr₂Ge₂Te₆ (30 – 41 K)³³ and those in other 2D FM materials,^{30,34–37} such as FeCl₂ (109 K),³⁸ NiBr₂ (132 K),³⁸ FeI₂ (42 K),³⁸ and Fe₃GeTe₂ (130 K).³⁵

We investigated the electronic and magnetic properties of monolayer NbMnO₂Cl₄ under various biaxial strains. As shown in Fig. 3, applying a compressive strain of -4% opens a bandgap of 0.06 eV in the spin-up channel, indicating a transition from HM to half-semiconducting (HS) behavior. The semiconducting electronic properties have been further confirmed using the HSE06 method (Fig. S9). Increasing the compressive strain further results in a continuous widening of the bandgap. Similarly, tensile strain induces an analogous

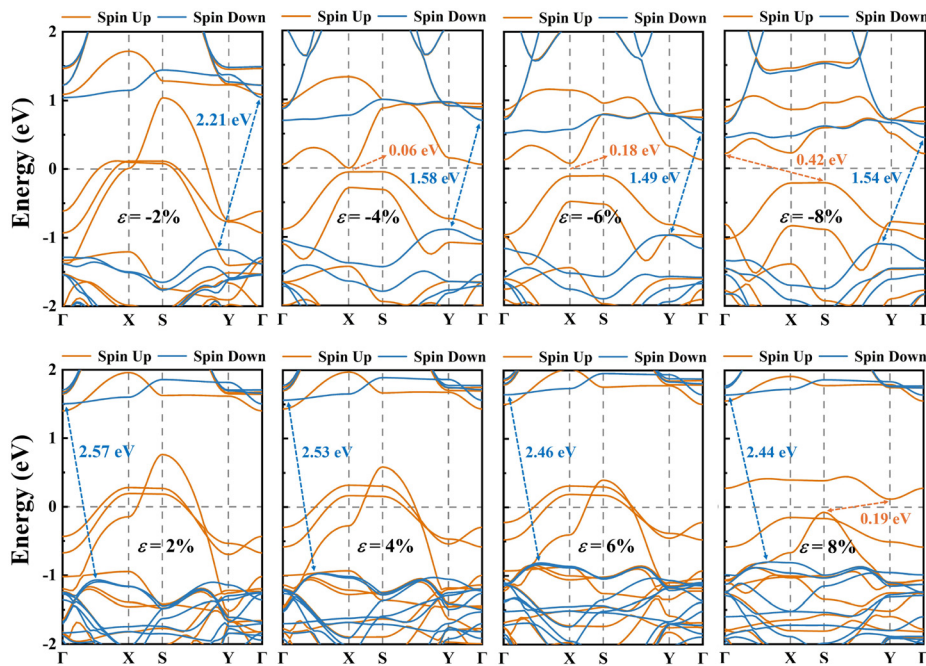


FIG. 3. Spin-polarized band structures of monolayer NbMnO₂Cl₄ calculated with $U = 5$ eV for Mn atom and $U = 3$ eV for Nb atom at different in-plane biaxial strain. The Fermi level is set to zero. Orange and blue lines indicate spin-up and spin-down states, respectively.

phase transition, opening a bandgap of 0.19 eV in the spin-up channel at 8% strain. In the spin-down channel, compressive strain reduces the bandgap, whereas tensile strain enlarges it.

Crystal structures of monolayer NbMnO₂Cl₄ under different strains are illustrated in Fig. S10. Compressive strain primarily induces off-center displacements of Mn and Nb atoms within their respective [MnO₂Cl₄] and [NbO₂Cl₄] octahedra along the b -axis, while tensile strain primarily drives a displacement of the Mn atom within its octahedron. Given that the electronic states near the Fermi level mainly originate from Mn 3d orbitals, we plotted the band structures projected onto the Mn 3d orbitals under varying strain conditions [Figs. 4(a)–4(d)], combined with the corresponding crystal structures to elucidate the mechanism underlying the HM-to-HS transition. With increasing compressive strain, the [MnO₂Cl₄] octahedron becomes increasingly

distorted, which raises the energy level of the $d_{x^2-y^2}$ orbital. As a result, the four Mn 3d electrons fully occupy the lower-energy d orbitals, leading to a semiconducting electronic structure. Similarly, tensile strain also distorts the [MnO₂Cl₄] octahedron, but in a different manner. This distortion leads to a rearrangement of the Mn 3d orbital energy levels, with the d_{xy} orbital shifting to the highest energy. When the tensile strain reaches 8%, the d_{xy} orbital becomes unoccupied, while the other four lower-energy orbitals are fully occupied, again resulting in a semiconducting state. These results underscore the crucial role of spin-lattice interactions in driving strain-induced phase transitions.

Due to the off-center displacement of Mn atoms, spontaneous polarization emerges when compressive strain exceeds -4% or tensile strain surpasses 8%. Notably, the mechanism of spontaneous polarization in NbMnO₂Cl₄ differs from that in NbOX₂, where the polarization

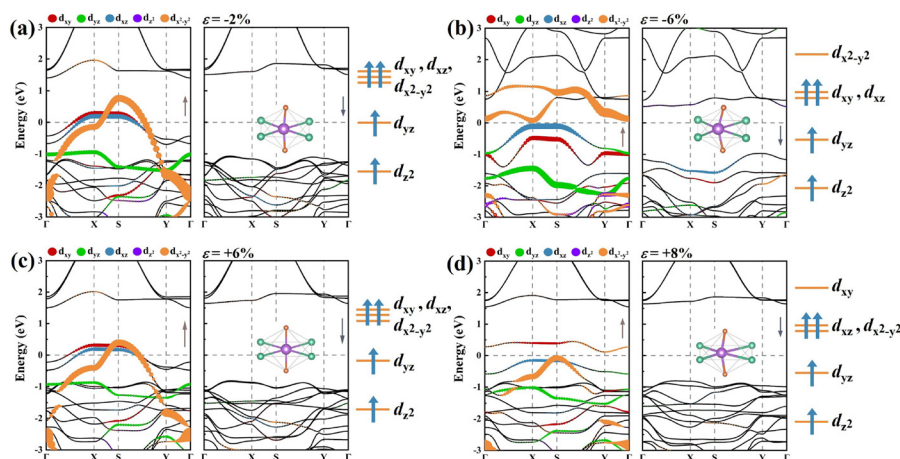


FIG. 4. Band structures of monolayer NbMnO₂Cl₄ projected onto Mn-3d orbitals, along with schematic diagrams of the Mn d -orbital energy levels under varying in-plane biaxial strain: (a) $\epsilon = -2\%$, (b) -6% , (c) 6% , and (d) 8% . Insets show the octahedral coordination environments around the Mn atom.

is primarily driven by the displacement of Nb atoms. In contrast, in $\text{NbMnO}_2\text{Cl}_4$, Mn atoms gradually deviate from their centrosymmetric positions as tensile strain increases, whereas both Nb and Mn atoms show progressive off-centering under compressive strain. This distortion enhances the polarization along the b direction (Fig. S11 and Table S2) and increasing the FE switching barrier [Fig. 5(a)].

To elucidate the polarization direction under biaxial strain, we performed phonon spectrum calculations for the nonpolar structures. Two phonon branches exhibit imaginary frequencies throughout the Brillouin zone, with the strongest instabilities located at the Γ and Y points [Figs. S12(a) and S12(b)]. The corresponding energy profiles of the distortion modes demonstrated that the Γ_2 mode dominates the polarity at -4% strain, while the Γ_1 mode is dominant at -6% strain, both associated with significantly deeper energy wells [Figs. S12(c) and S12(d)]. The resulting polar ground states exhibit a polarization direction along the b axis.

Berry phase calculations yield an in-plane polarization of 0.16×10^{-9} C/m along the b direction at -6% strain, with a moderate FE switching barrier of 0.08 eV. Notably, the switching pathway via an antiferroelectric phase exhibits a significantly higher energy barrier of 2.5 eV/f.u. (Fig. S13), indicating that $\text{NbMnO}_2\text{Cl}_4$ strongly favors a direct FE switching mechanism, without involving any intermediate antipolar configurations. The polarization is comparable to that of other well-studied 2D materials,^{13,39–52} such as SnS ,³⁹ VOI_2 ,⁴⁰ $\text{Bi}_2\text{O}_2\text{Se}$,⁴¹ $\gamma\text{-AlOOH}$,⁴² and $1\text{T}'\text{CrCoS}_4$.⁴³

Additionally, we systematically explored the magnetic ground state, MAE, and magnetic transition temperature of monolayer $\text{NbMnO}_2\text{Cl}_4$ under biaxial strain. Total energy calculations of four magnetic configurations relative to the FM state indicate that $\text{NbMnO}_2\text{Cl}_4$ exhibits a FM ground state in the strain range of -4%

to 6% , while outside this range, it possesses an AFM state [Fig. 5(b)]. As depicted in Figs. 5(c) and S14, compressive strain between -4% and -8% favors in-plane spin alignment, with the MAE reaching $670 \mu\text{eV}$ per Mn atom at -8% strain. Under tensile strain, the easy magnetization axis behavior is more complex: for tensile strain between 2% and 4% , the easy axis lies in the plane; at 6% strain, it switches to the out-of-plane direction but reverts back to in-plane alignment when the strain increases to 8% . The magnetic transition temperature (T_c/T_N) of $\text{NbMnO}_2\text{Cl}_4$ remains above room temperature for tensile strains exceeding 2% [Figs. 5(d) and S15], with T_N notably reaching as high as 605 K at 8% strain. Such strain conditions can be experimentally realized via epitaxial growth on substrates with varying lattice constants, where lattice mismatch induces the strain (Table S3).

In summary, we employed first-principles calculations to investigate the ferroic properties of monolayer $\text{NbMnO}_2\text{Cl}_4$. The intrinsic structure is found to be a HM–FM material, exhibiting a large MAE of $225 \mu\text{eV}$ per Mn atom and a high T_c of 285 K. By applying biaxial strain, the system undergoes a series of strain-driven phase transitions. Under compressive strain, a two-step transformation is found: from HM–FM to a FE–FM phase and then to a FE–AFM state. In contrast, tensile strain induces a direct HM–FM to FE–AFM transition. These ferroic phase transitions originate from strain-induced distortion of the $[\text{MnO}_2\text{Cl}_4]$ octahedra that alter magnetic exchange interactions and polarization behavior. Furthermore, the magnetic and FE properties of monolayer $\text{NbMnO}_2\text{Cl}_4$ are highly tunable under biaxial strain. At 6% tensile strain, the T_c rises markedly to 540 K, indicating enhanced thermal stability of magnetic ordering. Under compressive strain, the easy axis of magnetization undergoes a transition from out-of-plane to in-plane, accompanied by a significant increase in MAE. In particular, the MAE reaches $670 \mu\text{eV}$ per Mn atom at -8% compressive strain.

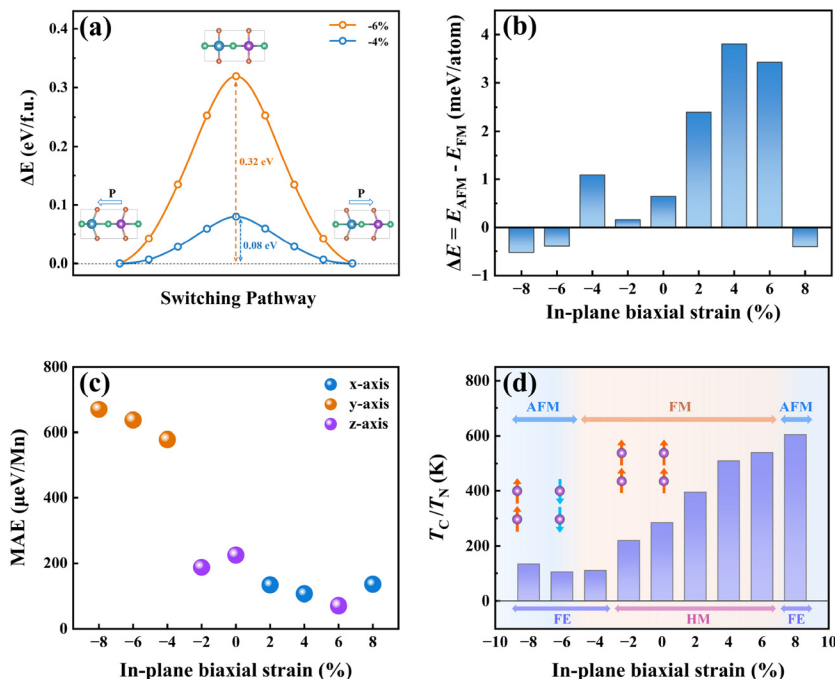


FIG. 5. (a) Energy profile along the FE switching pathway for monolayer $\text{NbMnO}_2\text{Cl}_4$ under $\epsilon = -4\%$ and -6% , showing the relative energy with respect to the initial FE phase. Insets depict the crystal structures of the initial, transition, and final states during polarization reversal. Hollow cyan arrows indicate the polarization directions. (b) The energy difference between FM and AFM states, (c) MAE, and (d) Curie/Néel temperature (T_c/T_N) as a function of in-plane biaxial strain.

These results underscore the coexistence and tunable coupling of magnetic and FE orders in monolayer NbMnO₂Cl₄, highlighting its potential as a versatile platform for next-generation multifunctional devices.

See the [supplementary material](#) for computational details, the origin of negative Poisson's ratios, phonon spectra, AIMD simulations, Young's modulus and Poisson's ratio, the effect of the Hubbard U parameter applied to Nb atoms on the crystal structure, PDOS and HSE06-level band structure, angular-dependent MAE and temperature-dependent specific heat under various strains, and the electronic polarization at strains of -6% , -8% , and 8% .

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yuntao Jie: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Software (equal); Writing – original draft (equal). **Jingyan Chen:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Software (equal). **Meiling Xu:** Funding acquisition (equal); Supervision (lead); Writing – review & editing (equal). **Yinwei Li:** Funding acquisition (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

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